

Selective Memory: How AI Retrieval Systems May Favor Certain Voices in Group Conversations

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Abstract

AI-powered retrieval-augmented generation (RAG) systems are used in collaborative workplace settings to integrate group conversations and support decision-making. However, the underlying retrieval mechanisms may favor certain communication styles over others, potentially reinforcing existing power dynamics. This project investigates whether modern RAG systems with neural embeddings and large language models exhibit selection bias based on the linguistic characteristics of the speakers. We implement a complete RAG pipeline using Sentence-BERT embeddings, ChromaDB vector database, and Claude Sonnet 4.6 for answer generation, while also comparing TF-IDF and Dense retrieval baselines. Using local JSONL trace logging, we analyze which speakers' messages are retrieved most frequently across 25 group conversations (1,601 messages) and examine selection patterns using correlation analysis and logistic regression over linguistic features such as verbosity, hedging, and assertiveness. Finally, we conduct a human study to measure whether LLM-generated summaries based on biased retrieval affect perceptions of speaker importance and expertise. This work contributes empirical evidence of systematic bias in modern RAG systems and demonstrates its potential social impact in collaborative settings.

1 Introduction

The modern workplace increasingly relies on AI tools to manage information overload in collaborative settings. Meeting transcription services, chat summarization tools, and retrieval-augmented generation (RAG) systems promise to surface the most relevant contributions from lengthy group discussions. These systems act as computational memory, deciding whose ideas are brought forward and whose are left behind.

However, retrieval algorithms do not operate in the absence of social context. Research in sociolin-

guistics demonstrates that language style significantly influences perceptions of authority, confidence, and credibility (Giles and Ogay, 2007; Bell, 1984). Some speakers communicate curtly and directly, while others provide comprehensive, nuanced explanations. Some use hedging language such as "maybe" and "I think", while others make assertive statements. If retrieval systems systematically favor certain linguistic patterns, they may amplify some voices while suppressing others – even when all participants contribute equally valuable insights.

This project addresses a critical gap at the intersection of natural language processing, human-computer interaction, and fairness in AI systems. While previous work has examined bias in model outputs and training data (Bender et al., 2021), fewer studies have explored how the *selection* of past content in RAG systems shapes social perceptions in group settings. We investigate three research questions:

RQ1: When a RAG system with neural embeddings retrieves messages from a group conversation, whose messages does it tend to select most often?

RQ2: Do writing style differences, such as verbosity, hedging, or assertiveness, systematically influence which messages are retrieved by similarity algorithms?

RQ3: Do LLM-generated summaries based on selectively retrieved messages lead readers to perceive certain speakers as more important or knowledgeable than they actually were?

We implement a complete RAG-style pipeline using Sentence-BERT embeddings, ChromaDB vector retrieval, TF-IDF and Dense retrieval baselines, and Claude Sonnet 4.6 for answer generation. Using local JSONL trace logging, we log retrieval decisions across 25 group conversations, enabling detailed analysis of selection patterns and their correlation with linguistic features.

This work makes several contributions: (1) we

provide empirical evidence of selection bias in modern retrieval systems, (2) we identify specific linguistic features that predict retrieval likelihood through correlation analysis and logistic regression, (3) we measure whether retrieval bias translates into human perception bias through a controlled study, and (4) we release our complete implementation, dataset, and analysis tools to enable future research on fairness in retrieval systems.

2 Related Work

2.1 Bias in NLP Systems

Recent work has identified various forms of bias in language models and NLP systems. [Bender et al. \(2021\)](#) highlight how training data composition and scale can encode societal biases, leading models to exhibit discrimination based on gender, race, and other protected attributes. [Bolukbasi et al. \(2016\)](#) demonstrate gender bias in word embeddings, showing that semantic representations reflect and amplify stereotypes present in training corpora.

However, most bias research focuses on model outputs (generated text) or learned representations (embeddings). Our work examines bias in the *retrieval* phase of RAG systems, a distinct but equally important source of potential unfairness.

2.2 Communication Style and Perception

Communication Accommodation Theory ([Giles and Ogay, 2007](#)) posits that language style significantly affects how speakers are perceived in terms of competence, authority, and social status. [Bell \(1984\)](#) demonstrates that speakers adjust their language style based on their audience, and these adjustments carry social meaning.

In workplace settings, linguistic patterns correlate with perceived leadership and influence. Verbose, elaborative communication may signal thoughtfulness but can also be dismissed as lacking confidence. Conversely, terse statements may convey authority but omit important nuance. Our work investigates whether retrieval algorithms encode preferences for specific communication styles, potentially disadvantaging certain demographic or professional groups.

2.3 Retrieval-Augmented Generation

RAG systems ([Lewis et al., 2020](#)) have become increasingly popular for question-answering and

summarization tasks. These systems retrieve relevant passages from a knowledge base and use them as context for generation. While prior work evaluates RAG systems on accuracy and relevance, we examine their fairness properties, specifically, whether they provide equitable representation of different speakers’ contributions.

2.4 Fairness in Retrieval-Augmented Generation

Recent work has begun to examine fairness as a system-level concern in RAG pipelines. [Wu et al. \(2024\)](#) evaluate fairness in RAG systems across demographic attributes and find that disparities can persist in both retrieval and generation stages, even when systems improve utility. [Avula et al. \(2025\)](#) similarly argue that fairness in RAG should be measured across the retrieval-generation pipeline, showing that disparities in retrieved evidence can propagate into generated outputs. [Kim et al. \(2025\)](#) focus on the embedding component of RAG systems, demonstrating that the embedder, corpus, and generator can each contribute to downstream bias.

Our work builds on this emerging literature but shifts the unit of fairness from demographic attributes or documents to *speaker visibility* in group conversations. Rather than asking only whether a RAG system produces fair answers, we examine whether its retrieval stage disproportionately preserves certain speakers’ contributions and communication styles, and whether those retrieval patterns shape downstream human perceptions of importance, expertise, and trust.

2.5 Visibility in Collaborative Systems

HCI research demonstrates that technology design affects participation and visibility in group settings ([DiMicco et al., 2004](#)). Systems that surface certain contributions more prominently can reinforce status hierarchies. Our work extends this line of inquiry to AI-mediated group memory, examining how retrieval algorithms shape whose ideas remain visible over time.

3 Methodology

3.1 Data Collection and Preparation

We utilize a dataset of 25 realistic group conversations containing 1,601 messages from 7 participants. Each message includes speaker ID, timestamp, and text content. The conversations revolve around collaborative project planning, providing

realistic examples of multi-turn, multi-speaker discussions with diverse communication styles.

Conversations are stored in CSV format with the schema: `conv_id`, `msg_id`, `speaker_id`, `timestamp`, `text`. This format allows reproducible analysis and easy extension to additional datasets.

3.2 RAG Pipeline Implementation

We implement a complete modern RAG system using multiple retrieval methods. Figure 1 summarizes the overall workflow from conversation preprocessing and retrieval to bias analysis, summary generation, and human evaluation.

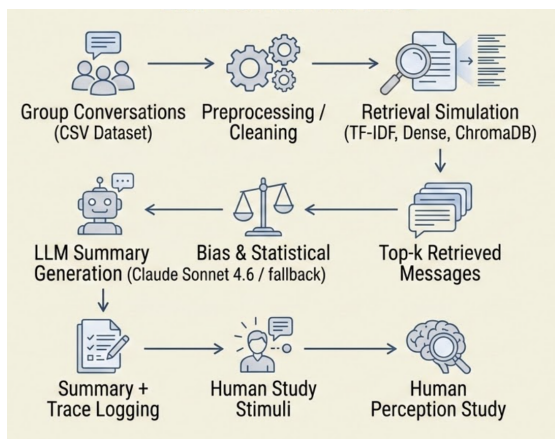


Figure 1: Overview of the project pipeline, from conversation preprocessing and retrieval simulation to statistical bias analysis, summary generation, and human perception evaluation.

Step 1 - Embedding Generation: We use Sentence-BERT (Reimers and Gurevych, 2019) (all-MiniLM-L6-v2 model) to generate 384-dimensional dense embeddings for each message. This model captures semantic meaning beyond simple keyword matching, enabling retrieval of contextually similar messages even when they use different vocabulary.

Baseline Retrieval: To contextualize the behavior of neural embeddings, we implement a TF-IDF retrieval baseline using cosine similarity over sparse vector representations. This allows us to compare representation imbalance across classical lexical retrieval and dense semantic retrieval methods. Both retrieval methods use identical top-*k* selection and prompt templates to ensure comparability.

Step 2 - Vector Database Indexing: All message embeddings are indexed in ChromaDB, a specialized vector database optimized for similarity

search. Each message is stored with metadata including speaker ID, timestamp, and conversation ID, enabling filtered retrieval and detailed logging.

Step 3 - Retrieval and Summary Generation: We implement a complete RAG-style pipeline with the following components:

- **Retriever:** TF-IDF, Dense retrieval, and ChromaDB retrieval are used to find the top-5 most relevant messages
- **LLM:** Claude Sonnet 4.6 generates natural language summaries from retrieved context
- **Prompt Template:** Instructs the LLM to cite speakers when referencing their contributions

Step 4 - Trace Logging: We enable detailed local logging using JSONL files, which record:

- Every query and retrieved messages
- Speaker IDs for each retrieved message
- Similarity scores and rankings
- LLM-generated answers with speaker citations

For each conversation, we ask five standardized questions:

- What is the main topic of discussion?
- What are the key points being discussed?
- What decisions were made?
- What concerns were raised?
- What is the recommended approach?

Each query retrieves the top-5 most relevant messages. The trace logs create a complete audit trail, enabling detailed analysis of which speakers are selected at each step. This generates 125 logged retrieval events (25 conversations × 5 questions) with full traceability.

3.3 Linguistic Feature Extraction

To understand *why* certain speakers are favored, we extract 15+ linguistic features from each message:

Verbosity Metrics:

- Word count, character count
- Average word length
- Words per sentence

Hedging and Uncertainty:

- Hedge word count and ratio (e.g., "maybe," "perhaps," "I think")
- Modal verbs expressing possibility

Assertiveness:

- Assertive markers (e.g., "definitely," "clearly," "must")
- Declarative vs. interrogative sentences
- Confidence score (assertiveness - hedging)

Discourse Features:

- Question frequency
- Politeness markers (e.g., "please," "thank you")
- Meta-conversation phrases (e.g., "Quick question:")

We report speaker-level feature averages and Pearson correlations as descriptive analyses, and use message-level logistic regression for statistical testing.

3.4 Statistical Modeling of Retrieval Likelihood

Speaker-level correlations ($n=7$) provide limited statistical power. We therefore conduct message-level analysis to obtain more reliable estimates.

For each message i , we define a binary outcome variable:

$$Y_i = \begin{cases} 1 & \text{if message } i \text{ is retrieved in any query} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

We model the probability of retrieval using logistic regression:

$$P(Y_i = 1) = \sigma(\beta_0 + \beta_1 V_i + \beta_2 H_i + \beta_3 A_i + \dots) \quad (2)$$

where V_i , H_i , and A_i denote verbosity, hedging, and assertiveness features for message i .

This tests whether linguistic features significantly predict retrieval likelihood beyond participation volume. We fit both a full model using linguistic features and speaker identity, and a style-only model using linguistic features alone.

3.5 Bias Quantification

To evaluate whether the retrieval system disproportionately amplifies certain speakers, we compute multiple representation-aware metrics.

Participation Share: For each speaker s , we compute their proportion of total contributions in the dataset:

$$\text{ParticipationShare}_s = \frac{\text{TotalMessages}_s}{\text{TotalMessages}} \quad (3)$$

Retrieval Share: We compute the proportion of retrieved messages attributed to each speaker:

$$\text{RetrievalShare}_s = \frac{\text{TimesRetrieved}_s}{\text{TotalRetrievedMessages}} \quad (4)$$

Representation Ratio: To measure over- or under-representation, we compute:

$$\text{RepresentationRatio}_s = \frac{\text{RetrievalShare}_s}{\text{ParticipationShare}_s} \quad (5)$$

A value greater than 1 indicates that a speaker is over-represented in retrieval relative to their participation, while a value less than 1 indicates under-representation.

Inequality Metrics: To quantify concentration of algorithmic attention across speakers, we compute two complementary metrics.

First, we compute the Gini coefficient over retrieval shares, where higher values indicate greater concentration of retrieval attention across speakers.

Second, we report the maximum retrieval gap:

$$\max_s(\text{RetrievalShare}_s) - \min_s(\text{RetrievalShare}_s) \quad (6)$$

3.6 Human Perception Study

To test RQ3, we conduct a human study with 16 participants using Google Forms.

Study Design: Participants read AI-generated summaries (from the LLM-based RAG system) of group conversations. Summaries cite speakers when referencing their contributions (e.g., "Speaker A suggested Python, while Speaker C emphasized keeping things simple").

Participants evaluate each summary on four dimensions:

- **Contribution:** Which speaker seemed to contribute the most to moving the discussion forward?

- **Knowledge:** Which speaker seemed the most knowledgeable or expert?
- **Trust:** Which speaker would you trust most when making decisions?
- **Balance:** How balanced did the summary seem across speakers? (5-point scale)

Each participant evaluated 25 summary blocks, producing 400 participant-block observations.

Analysis: We aggregate speaker selections and balance ratings to test whether selective summaries shape perceptions of speaker importance and expertise. We also analyze whether certain speakers are perceived as "leaders" or "experts" based on their prominence in AI-generated summaries, testing whether retrieval systems can artificially create or reinforce status hierarchies.

For reference, we compare perception patterns against each speaker’s participation share and retrieval prominence to assess divergence between algorithmic exposure and actual conversational activity.

3.7 Implementation and Tools

All code is implemented in Python 3.11 following the project methodology.

Core Technologies:

- **sentence-transformers:** Generate Sentence-BERT embeddings
- **chromadb:** Vector database for embedding storage and retrieval
- **anthropic:** Claude Sonnet 4.6 API for answer generation
- **scikit-learn:** Logistic regression modeling and preprocessing

Data Pipeline: Conversations are stored in CSV format with schema: conv_id, msg_id, speaker_id, timestamp, text. Python-based preprocessing handles data loading and validation.

Analysis Tools: NumPy and pandas are used for statistical analysis, and matplotlib for visualization. All linguistic feature extraction is implemented using regex and string operations.

Human Study: Responses are collected through Google Forms and exported as CSV for analysis.

Code, trace logs, and dataset are available at <https://github.com/jrawat2/SP26-CS498-Selective-Memory>.

4 Results

4.1 RQ1: Speaker Representation in Retrieval

We first evaluate whether retrieval systems represent speakers proportionally relative to their actual participation in the conversations. For each retriever, we compare each speaker’s participation share to their retrieval share, and compute representation-aware fairness metrics including representation ratio, Gini coefficient, and maximum retrieval-share gap.

As shown in Figure 2, retrieval share does not consistently match participation share across the three retrievers. Some speakers are over-represented in the retrieved set, while others are under-represented relative to how frequently they participated in the original conversations. This indicates that retrieval systems do not simply preserve conversational participation structure.

The representation-ratio comparison further shows that these imbalances differ by retrieval method. TF-IDF exhibits some speaker-level distortion, but the semantic retrievers show stronger shifts in representation for particular speakers. This pattern is also reflected in the Gini coefficient shown in Figure 3, which summarizes how unevenly retrieval attention is distributed across speakers. In our results, TF-IDF produces the lowest retrieval inequality, while Dense and Chroma produce higher inequality, suggesting that semantic retrieval concentrates retrieval attention more strongly on a smaller subset of speakers.

Together, these findings support RQ1: retrieval systems do not represent speakers proportionally, and semantic retrievers can amplify certain voices more strongly than the lexical baseline.

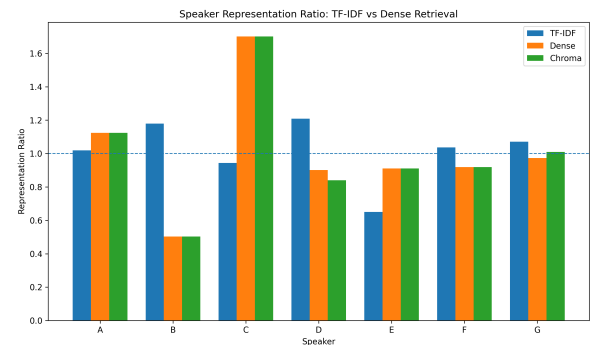


Figure 2: Speaker-level representation ratios across retrieval methods. Values above 1 indicate over-representation in retrieval relative to participation, while values below 1 indicate under-representation.

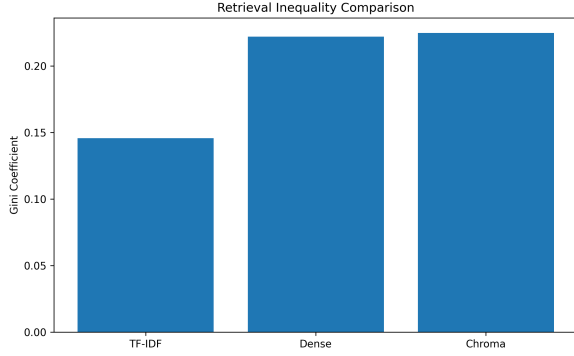


Figure 3: Retrieval inequality across methods measured by the Gini coefficient. Higher values indicate greater concentration of retrieval attention on a smaller subset of speakers.

4.2 RQ2: Linguistic Style and Retrieval Likelihood

We next examine whether linguistic style influences retrieval likelihood at the message level. Using speaker-level feature summaries, Pearson correlations, and message-level logistic regression, we evaluate whether retrieval is associated with stylistic properties such as verbosity, hedging, assertiveness, modal language, and formatting style.

Across the style-based analyses, longer messages were more likely to be retrieved, especially in the Dense and Chroma settings. More assertive and confidence-related language was also positively associated with retrieval. In contrast, modal-heavy or more qualified language was less likely to be retrieved, and uppercase-heavy or visually noisy text also showed a negative relationship with retrieval. More fragmented sentence structure further reduced retrieval likelihood in the semantic retrievers.

The logistic regression results reinforce these patterns. In the style-only models, linguistic features alone achieved 58.6% training-set accuracy for TF-IDF and 63.8% for both Dense and Chroma. As summarized in Table 1, for the semantic retrievers, message length in words had an odds ratio of approximately 2.43, assertive language had an odds ratio of approximately 1.77, while modal verbs and uppercase-heavy text had odds ratios of approximately 0.50 and 0.41 respectively. These values indicate that writing style meaningfully influences retrieval likelihood even when speaker identity is excluded.

These findings support RQ2: retrieval is not style-neutral, and linguistic style contributes meaningfully to which messages are surfaced by retrieval

Table 1: Full style-only regression coefficients for Chroma retrieval

Feature	Coefficient	Odds Ratio
Number of sentences	−0.925	0.396
Uppercase-heavy text	−0.902	0.406
Message length (words)	0.887	2.427
Words per sentence	−0.773	0.462
Modal verbs	−0.703	0.495
Message length (chars)	0.628	1.873
Assertive language	0.572	1.772
Assertive ratio	−0.355	0.701
Hedges	0.284	1.328
Hedge ratio	−0.223	0.800
Confidence score	0.201	1.223
Modal ratio	0.191	1.210
Has question	0.171	1.187
Number of questions	0.109	1.115
First-person pronouns	0.075	1.078
Meta-conversation phrases	0.073	1.076
Average word length	−0.036	0.965
Politeness markers	0.004	1.004
Question ratio	−0.001	0.999
Has exclamation	0.000	1.000
Number of digits	0.000	1.000

systems.

4.3 RQ3: Human Perception Effects

Finally, we examine whether selective summaries shape downstream human perception. Participants read AI-generated summaries and judged which speaker seemed most contributing, most knowledgeable, and most trustworthy, while also rating how balanced each summary appeared across speakers.

As shown in Figure 4, the human-study results show that participants did not perceive all speakers equally. Instead, judgments were concentrated on a small subset of speakers. Speaker C was most frequently selected as the top contributor (37.5%), the most knowledgeable speaker (33.75%), and the most trusted speaker (32.5%). Speakers B and F emerged as strong secondary figures, while Speakers D and E were rarely perceived as central. Speaker G was selected relatively often for knowledge, but less so for contribution and trust, suggesting a more informational than leadership-oriented role.

Balance ratings provide additional nuance. As shown in Figure 5, the mean balance rating was 3.47 out of 5, and the most common rating was 3, accounting for 52.25% of all responses. This suggests that participants did not view the summaries as extremely one-sided, but also did not perceive them as fully balanced across speakers.

Taken together, these results suggest that selec-

tive summaries shape human perception by making a small subset of speakers appear more important, knowledgeable, and trustworthy than others. This supports RQ3 and indicates that retrieval bias can propagate into downstream social judgments.

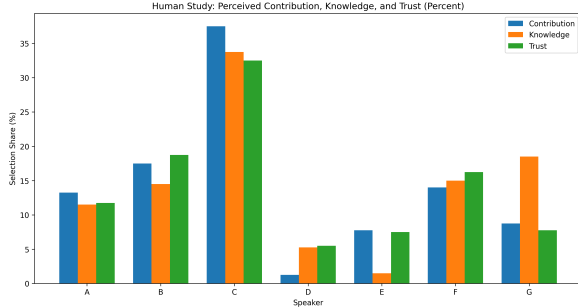


Figure 4: Human-study results showing perceived contribution, knowledge, and trust across speakers.

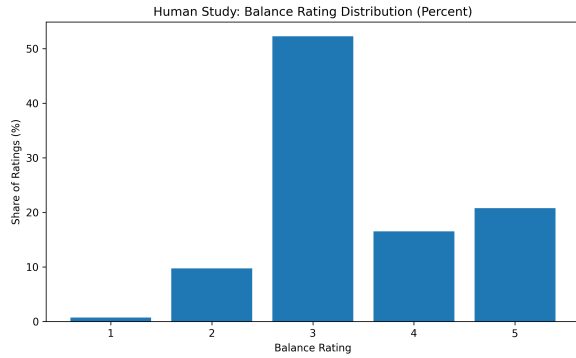


Figure 5: Human-study balance ratings across summaries.

5 Discussion

Our findings suggest that retrieval is not a neutral step in RAG systems. Across all three research questions, we observe that retrieval affects not only which messages are surfaced, but also which speakers become more visible and how those speakers are perceived in downstream summaries. In this sense, retrieval functions as a selective memory mechanism rather than a passive filtering step.

For RQ1, we found that retrieval systems did not represent speakers proportionally relative to their participation in the conversations. Although all three retrievers showed some imbalance, the semantic methods produced stronger speaker-level inequality than the TF-IDF baseline in our dataset. For RQ2, we found that retrieval likelihood was also shaped by linguistic style. Longer, more assertive, and more confident messages were more likely to be retrieved, while modal-heavy,

uppercase-heavy, or more fragmented messages were less likely to surface. Together, these results suggest that both speaker representation and message style influence what retrieval systems preserve.

For RQ3, the human-study results suggest that these retrieval patterns can propagate into downstream perception. Participants did not judge all speakers equally; instead, they concentrated their judgments of contribution, knowledge, and trust on a small subset of speakers, especially Speaker C. At the same time, the balance ratings were moderate rather than extreme, suggesting that selective summaries can shape perception even when they do not appear obviously one-sided. This makes the effect more subtle, but still socially meaningful.

These results connect to prior work in both NLP and HCI. Prior research has shown that AI systems can encode bias in model outputs and learned representations, while HCI work has shown that system design affects visibility and participation in collaborative settings. Our findings extend this literature by showing that bias can also emerge in the retrieval stage, which acts as a fairness bottleneck by determining which parts of a conversation remain visible for summarization and interpretation. More broadly, this work suggests that AI-supported organizational memory is selective rather than neutral, and that fairness in retrieval should be evaluated not only technically, but also in terms of downstream human perception.

6 Limitations and Ethical Considerations

This study is limited by the size and domain of the dataset (25 conversations, 7 participants), which may restrict generalizability to other organizational contexts. The retrieval analysis also relies on five standardized queries per conversation, which provide controlled comparisons across retrievers but may not capture the full range of real-world user prompts. Linguistic features are operationalized using surface-level markers and may not fully capture nuanced communication styles. Additionally, retrieval behavior may vary across embedding models and summary-generation models, so some findings may be model-dependent.

The human study is also modest in scale, with 16 participants and 25 summary blocks. As a result, the perception findings should be interpreted as evidence of meaningful associations rather than broad causal generalization. More broadly, the study fo-

cuses on perceived contribution, knowledge, trust, and balance, rather than longer-term behavioral consequences such as decision outcomes or workplace influence.

All conversations were anonymized prior to analysis. The human perception study was conducted using informed consent, and responses were analyzed only in aggregate form. Our goal is not to evaluate individual speakers, but to understand how system design choices may unintentionally shape visibility and perception in collaborative environments. Ethically, these findings matter because retrieval systems can influence whose voices appear most important in group settings. If such systems systematically favor certain speakers or communication styles, they may reinforce inequities in visibility, recognition, and perceived expertise.

7 Future Work

Several directions remain for extending this work. First, future studies should evaluate the framework on larger and more naturalistic datasets to test whether the observed retrieval and perception effects generalize beyond this course-scale setting. This includes applying the same analysis to other collaborative domains such as workplace meetings, online forums, and support chats.

Second, future work should compare a broader range of retrieval and generation configurations. While this project examined TF-IDF, Dense retrieval, and ChromaDB, additional retrievers, embedding models, and summary-generation settings could reveal whether certain architectures are more robust to representation imbalance than others. Similarly, varying retrieval depth or prompt design may help clarify how technical design choices influence downstream fairness.

Third, more extensive human evaluation would strengthen the findings. Larger participant pools and more diverse summary conditions could help test whether the effects we observed remain stable across populations and tasks. Future studies could also examine longer-term outcomes, such as whether selective summaries affect memory, decision-making, or perceived leadership over time.

Finally, future work should explore mitigation strategies. Potential directions include diversity-aware retrieval, fairness-aware ranking objectives, style-normalized retrieval representations, and interfaces that make retrieval selectivity more trans-

parent to users. These approaches would help move from bias detection toward the design of retrieval systems that are both useful and equitable.

8 Conclusion

This project investigated whether retrieval-augmented generation systems exhibit selective memory in multi-speaker conversations. Across three retrieval settings—TF-IDF, Dense retrieval, and ChromaDB—we found that retrieval does not represent speakers proportionally, that linguistic style meaningfully influences which messages are surfaced, and that selective summaries can shape downstream human judgments of contribution, knowledge, and trust.

Taken together, these findings show that retrieval is not a neutral intermediate step, but a socially consequential component of AI-supported summarization systems. By combining retrieval analysis, message-level statistical modeling, and human evaluation, this work highlights the need to audit retrieval systems not only for relevance, but also for representation fairness and downstream perceptual effects in collaborative environments.

9 Contributions

This project was completed collaboratively by Jyoti Rawat, Sandra John, and Bhavyaa Chauhan. All major methodological decisions, experimental design choices, analytical interpretations, and revisions to the paper were discussed jointly. At the same time, each team member took primary responsibility for specific parts of the implementation and analysis pipeline.

Jyoti Rawat led dataset preparation and preprocessing, including organizing the conversation data into a structured analysis pipeline. She also contributed to embedding generation, ChromaDB indexing, and linguistic feature extraction, including features related to verbosity, hedging, assertiveness, and related stylistic markers used in the regression analysis.

Bhavyaa Chauhan led retrieval-system implementation, including the TF-IDF, Dense, and Chroma retrieval pipelines. She also contributed to prompt design, summary-generation setup, and retrieval logging, ensuring that outputs from the retrieval stage were consistently captured for downstream analysis and human evaluation.

Sandra John led the quantitative analysis components of the project, including statistical model-

ing, fairness analysis, and visualization. Her contributions included computing participation and retrieval shares, representation ratios, Gini coefficients, logistic regression analyses, and the aggregation and interpretation of human-study results.

All three authors contributed to system validation, interpretation of findings, paper writing and revision, and presentation preparation. The final paper and presentation reflect shared discussion and iterative feedback across the full team.

References

- Sandeep Avula, Chia-Jung Lee, Rongting Zhang, and Vanessa Murdock. 2025. Measuring the fairness gap between retrieval and generation in rag systems using a cognitive complexity framework. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- Allan Bell. 1984. Language style as audience design. *Language in Society*, 13(2):145–204.
- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623.
- Tolga Bolukbasi, Kai-Wei Chang, James Y Zou, Venkatesh Saligrama, and Adam T Kalai. 2016. Man is to computer programmer as woman is to home-maker? debiasing word embeddings. In *Advances in Neural Information Processing Systems*, volume 29.
- Joan M DiMicco, Katherine J Hollenbach, Alyssa Pandolfo, and Walter Bender. 2004. Influencing group participation with a shared display. In *Proceedings of the 2004 ACM Conference on Computer Supported Cooperative Work*, pages 614–623.
- Howard Giles and Tania Ogay. 2007. Communication accommodation theory. *Explaining Communication: Contemporary Theories and Exemplars*, pages 293–310.
- Taeyoun Kim, Jacob Mitchell Springer, Aditi Raghunathan, and Maarten Sap. 2025. Mitigating bias in rag: Controlling the embedder. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 18999–19024.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992.
- Xuyang Wu, Shuowei Li, Hsin-Tai Wu, Zhiqiang Tao, and Yi Fang. 2024. Does rag introduce unfairness in llms? evaluating fairness in retrieval-augmented generation systems. *arXiv preprint arXiv:2409.19804*.